Missing Values

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Missing Values

In statistics, missing data, or missing values, occur when no data value is stored for the variable in an observation. Missing data are a common occurrence and can have a significant effect on the conclusions that can be drawn from the data.

- Data can be missing **at random** or **not at random**
- In **metabolomics** missing values arise when, for a sample, it was not possible to get a specific intensity for a feature/metabolite

Notes

In metabolomics missing values pop up for different reasons

1. There was an error during preprocessing (targeted or untargeted)

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2. For a sample the concentration of a metabolite was really low
3. A metabolite was "absent" in a sample I'll get a missing value (*eg. red pigment in a an extract of white grape*)

1,2 & 3 are **random** or **not random** ?

Analytical Absence

Having a **Missing** is **not** equivalent of having **zero** concentration!

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Where can we get the detection limit ?

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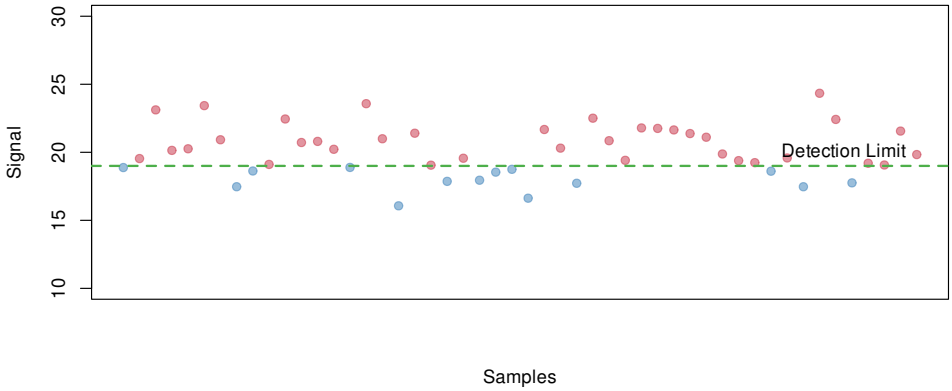
- Each analytical method is characterized by a well defined *detection limit*
- Every concentration between zero and the detection limit will not be detected
- We never can be 100% sure that something is absent from my sample ...



Where can we get the detection limit ?

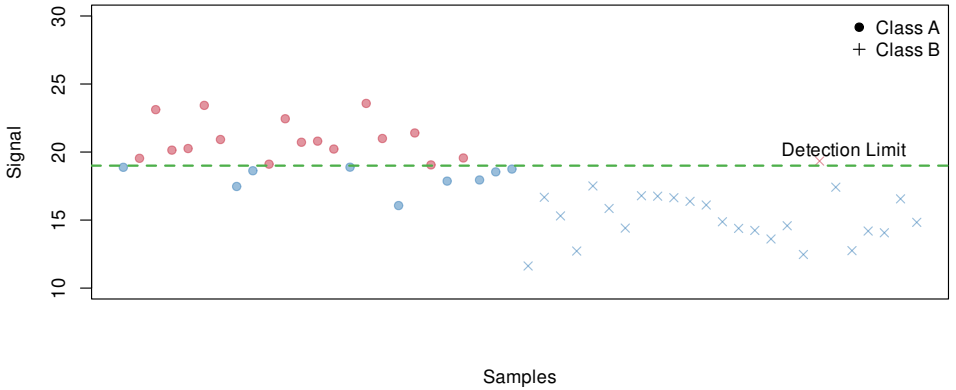
Scenario 1: 1 class

Signal close to the detection limit



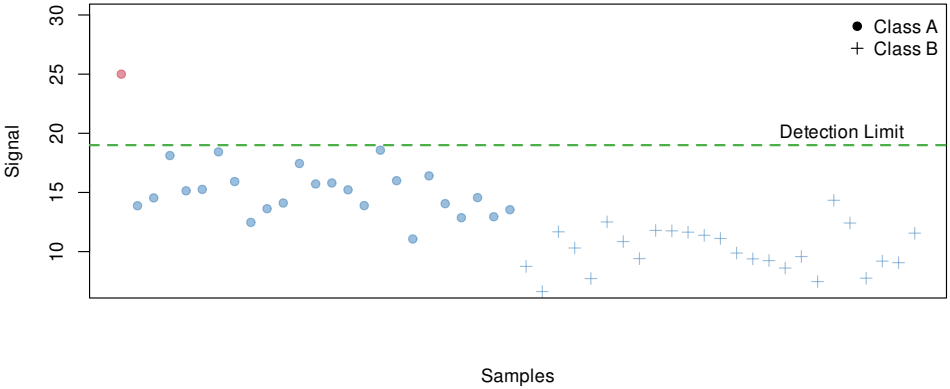
Missing values are “randomly” popping-up

Scenario 2: 2 class biomarker



Missing values are not "randomly" popping-up

Scenario 3: 2 class ??



Your turn



What would you propose to do in the previous three cases?

Dealing with missing values

- Use statistical methods able to handle missing data
- Impute them put a reasonable number with variability
- Remove features with too many NAs (scenario 3)

How many NAs are acceptable **depends on the allocation of the samples to the factors of the study**

Imputation

Imputation is the process of substituting a missing value with a reasonable number:

1. multivariate imputation uses the value of samples which are close in the multivariate space to select a good number (`missMDA` package, KNN imputation, ...). This works well if data are missing at random ...

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2. a reasonable number can be chosen on the bases of analytical considerations (e.g. a random number between zero and the detection limit)
3. the imputation strategy should be not "aware" of the design of the study

Further Observations

- It is “easier” to handle missing values randomly distributed
- Use **domain specific knowledge** (e.g analytical - LOD) inject new knowledge in the data analysis pipeline!
- **Quality Check**: try different forms of imputation: are the outcomes sensitive to that?